Forecast weights and forecast quantiles

Code (`or-joint-update-filter.R`, step 4):

```
wF <- as.vector(wPrior * piPrev); wF <- wF / sum(wF)
if (forecastQuantiles)
  for (a in 1:d) hist$fq[, t, a] <- mix_quantile(qs5, wF, muF[, a],
                                          sqrt(sigF[, mi(a, a, d)]))
```

Notation

Components are indexed by g (active τ grid point), h (history group) and k (disturbance scenario at step t).

- $\pi_g = p(\tau_g \mid y_{1:t-1})$: the τ posterior after step $t-1$ (`piPrev`, normalised `exp(qs)`).
- $\omega_{gh} = p(h \mid \tau_g, y_{1:t-1})$: history-group weight within grid point g, $\sum_h \omega_{gh} = 1$ (`prevW`).
- $\beta_k = p(k)$: prior probability of scenario k at step t (`dist.p[, t]`).
- $m_{ghk} \in \mathbb{R}^d$, $S_{ghk} \in \mathbb{R}^{d\times d}$: forecast mean and covariance of component (g, h, k) (`muF`, `sigF`).

1. Forecast weights

The prior weight of each component (`wPrior`) is

$$\tilde{w}_{ghk} = \omega_{gh}\,\beta_k\,\delta_{ghk}, \qquad \delta_{ghk} = \begin{cases} 0 & \text{component dropped,} \\ 1 & \text{otherwise,} \end{cases}$$

where a component is dropped if its importance weights collapsed or S_{ghk} is singular, except that a grid point with every $\delta_{ghk} = 0$ has all of its δ_{ghk} reset to 1. Multiplying by the τ weight and normalising gives the forecast weights

$$w^F_{ghk} = \frac{\pi_g\,\tilde{w}_{ghk}}{\sum_{g',h',k'} \pi_{g'}\,\tilde{w}_{g'h'k'}}.$$

(`wPrior * piPrev` multiplies element $[g, h, k]$ by π_g, since g is the fastest-varying index of the $G \times H \times K_1$ array.) Without dropped components the denominator is 1, and $w^F_{ghk} = p(\tau_g, h, k \mid y_{1:t-1})$.

2. Forecast distribution

The one-step-ahead forecast is the Gaussian mixture

$$p(x_t \mid y_{1:t-1}) \approx \sum_{g,h,k} w^F_{ghk}\,\mathcal{N}(x_t;\, m_{ghk},\, S_{ghk})\,.$$

The marginal of a Gaussian mixture is the mixture of the component marginals, so state element a has forecast CDF

$$F_a(x) = \sum_{g,h,k} w^F_{ghk}\,\Phi\left(\frac{x - m_{ghk,a}}{\sqrt{S_{ghk,aa}}}\right).$$

3. Forecast quantiles

For each $p \in \{0.025,\ 0.25,\ 0.5,\ 0.75,\ 0.975\}$ (`qs5`), the stored quantile `hist$fq[, t, a]` is the root $q_{a,p}$ of

$$F_a(q_{a,p}) = p.$$

`mix_quantile` solves this numerically:

- components with $w^F_{ghk} \le 10^{-10} \sum w^F$ are dropped and the remaining weights renormalised;
- `uniroot` searches the bracket $\big[\min(m_{ghk,a} - 8s_{ghk,a}),\ \max(m_{ghk,a} + 8s_{ghk,a})\big]$, with $s_{ghk,a} = \sqrt{S_{ghk,aa}}$, on which F_a goes from ≈ 0 to ≈ 1 (so a root exists, and is unique since F_a is increasing);
- the tolerance is 10^{-6} times the bracket width.

These are quantiles of the forecast *before* y_t is assimilated. The analysis quantiles (`hist$q`) are computed the same way from the posterior weights and the Kalman-updated moments. The forecast quantiles are returned in the output but are not currently plotted by any of the experiment scripts.